COMP2511

Tute05

Agenda

- Observer Pattern
- Composite Pattern
- Factory Pattern



Observer Pattern



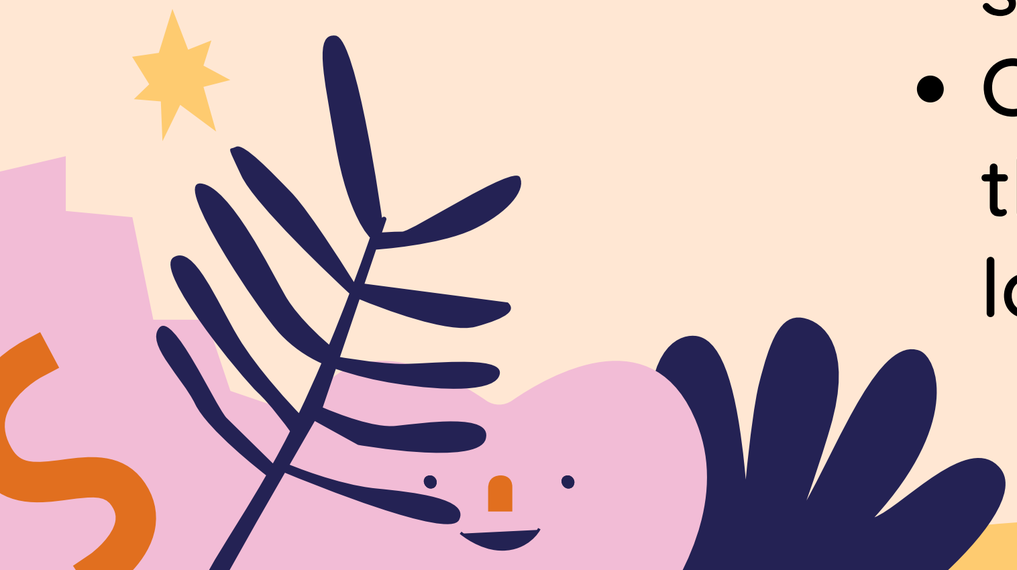


Observer Pattern

The common problem?

In our code, some observing objects in our code are waiting for some subject objects to update:

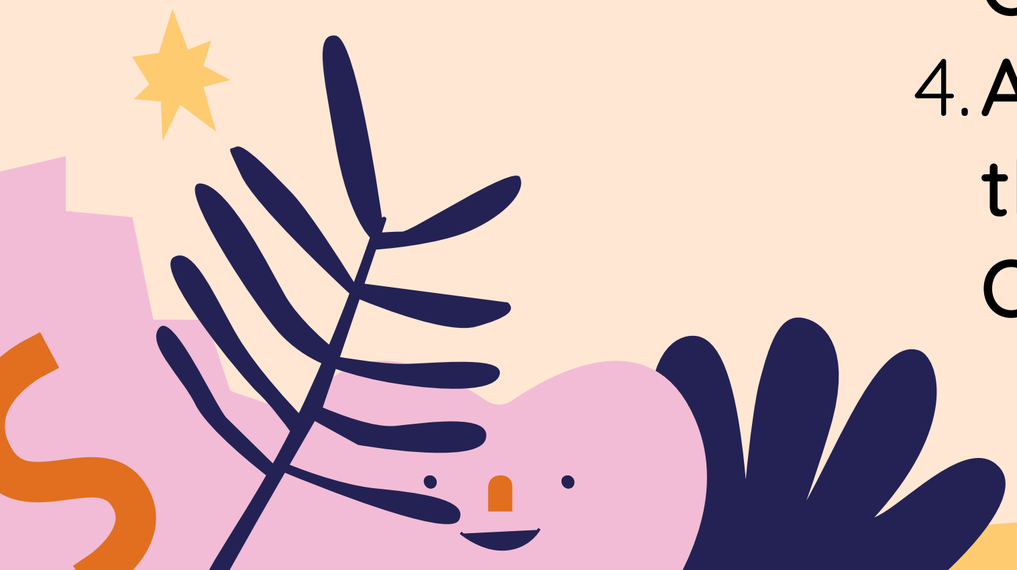
- Customers waiting for their local store to have PlayStation 5s in stock
- Children asking “are we there yet” to their parents every minute instead of looking out of the window to check



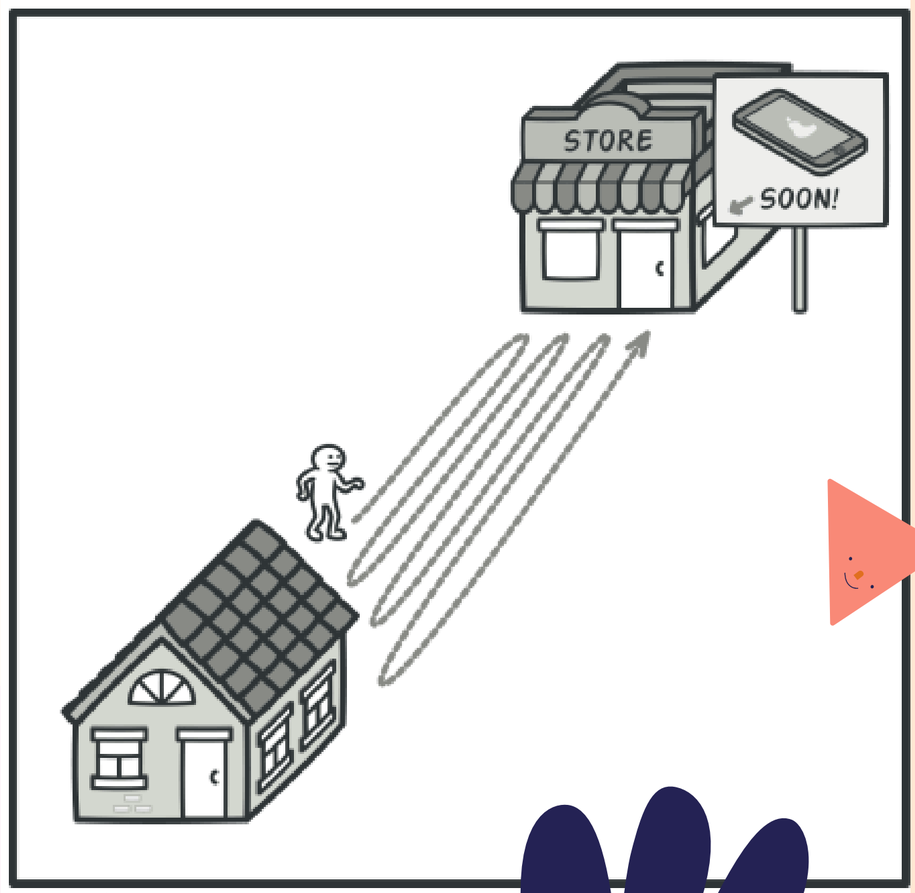
Observer Pattern

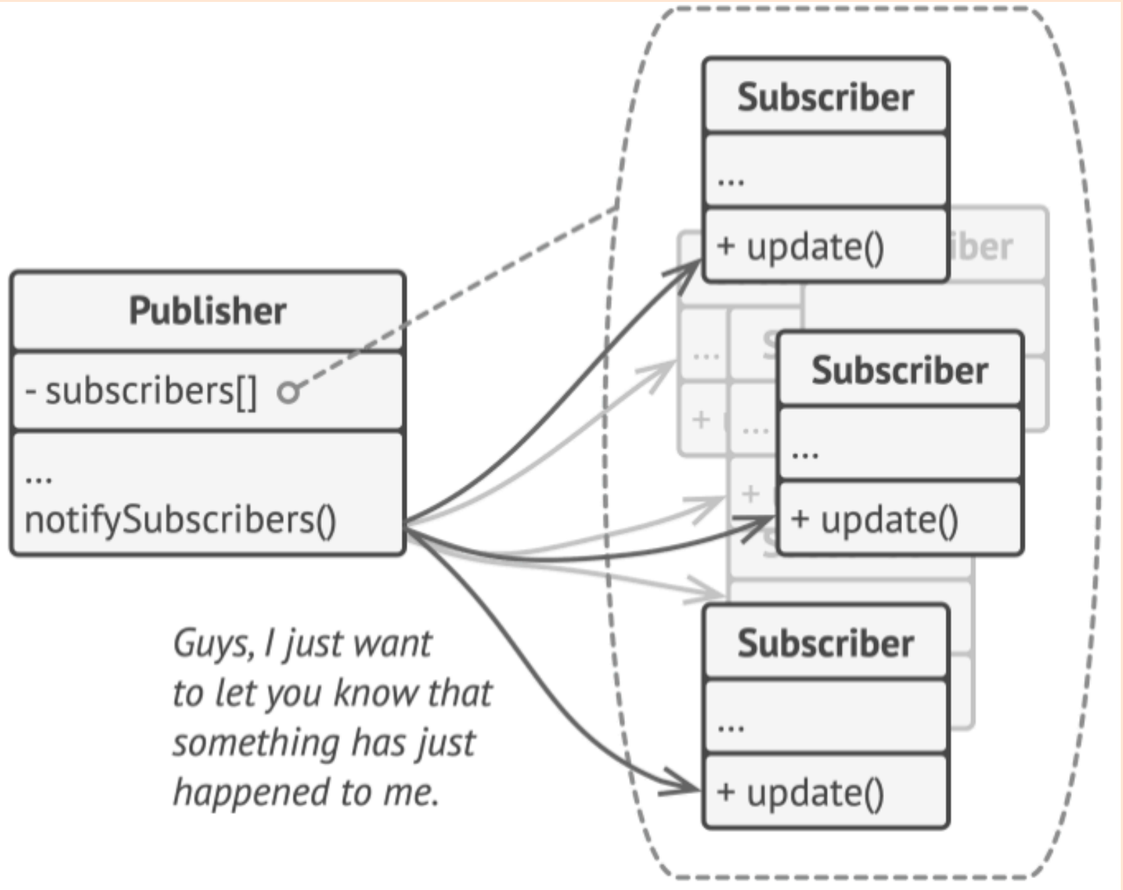
The common solution?

1. Identify the Subjects and Observers.
2. Keep a list of Observers subscribed to the Subject as an attribute.
3. Add an update() method to Observers.
4. Add a notify() method to Subject, that calls all of its subscribed Observer's update() methods.



Observer Pattern







Computer





Composite Pattern

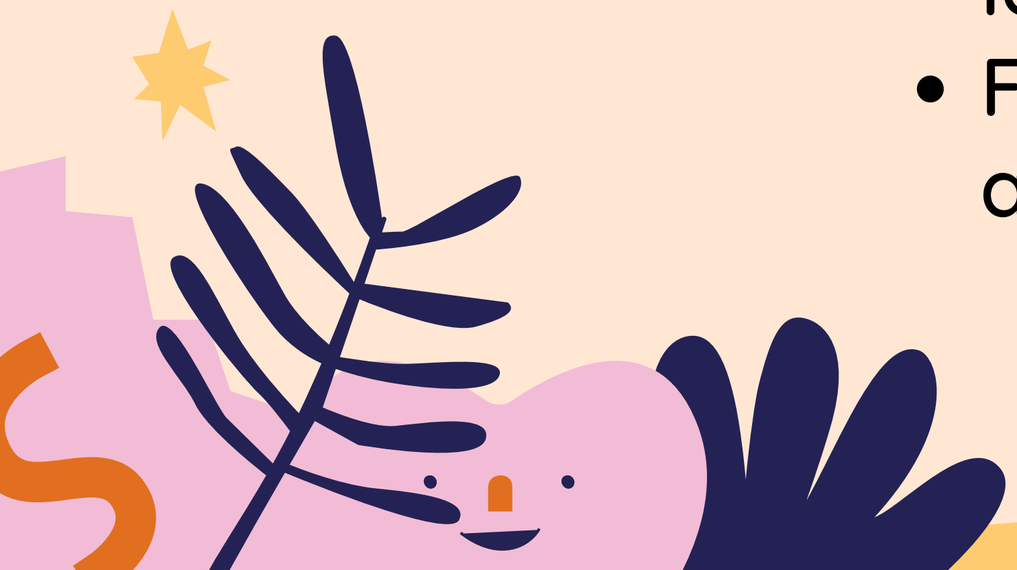


Composite Pattern

The common problem?

A model where we can treat individual objects (leaves) the same as a group of them (composites):

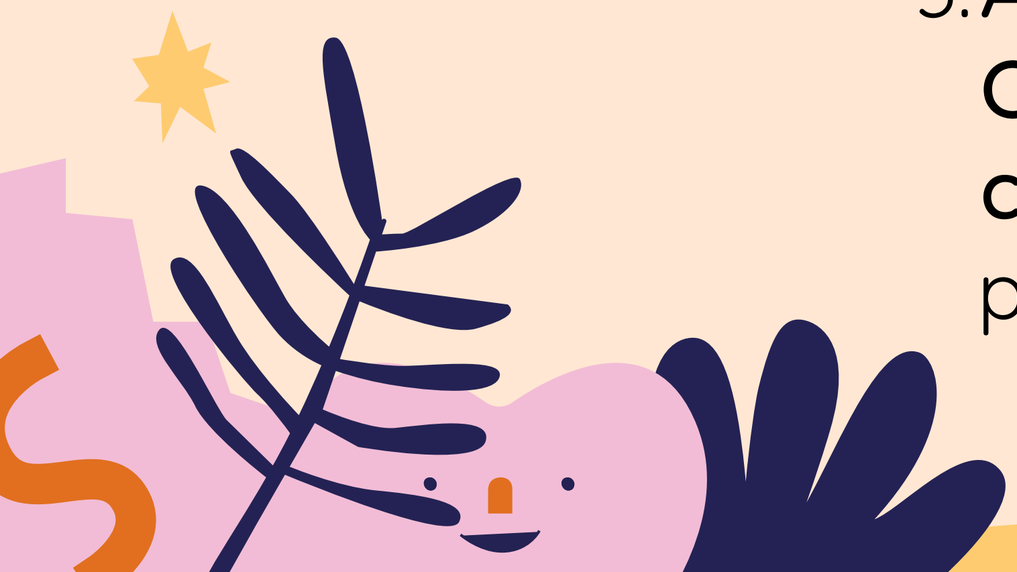
- Mathematical equation (numbers are leaves, operators are composites).
- File systems (files are leaves, folders are composites)



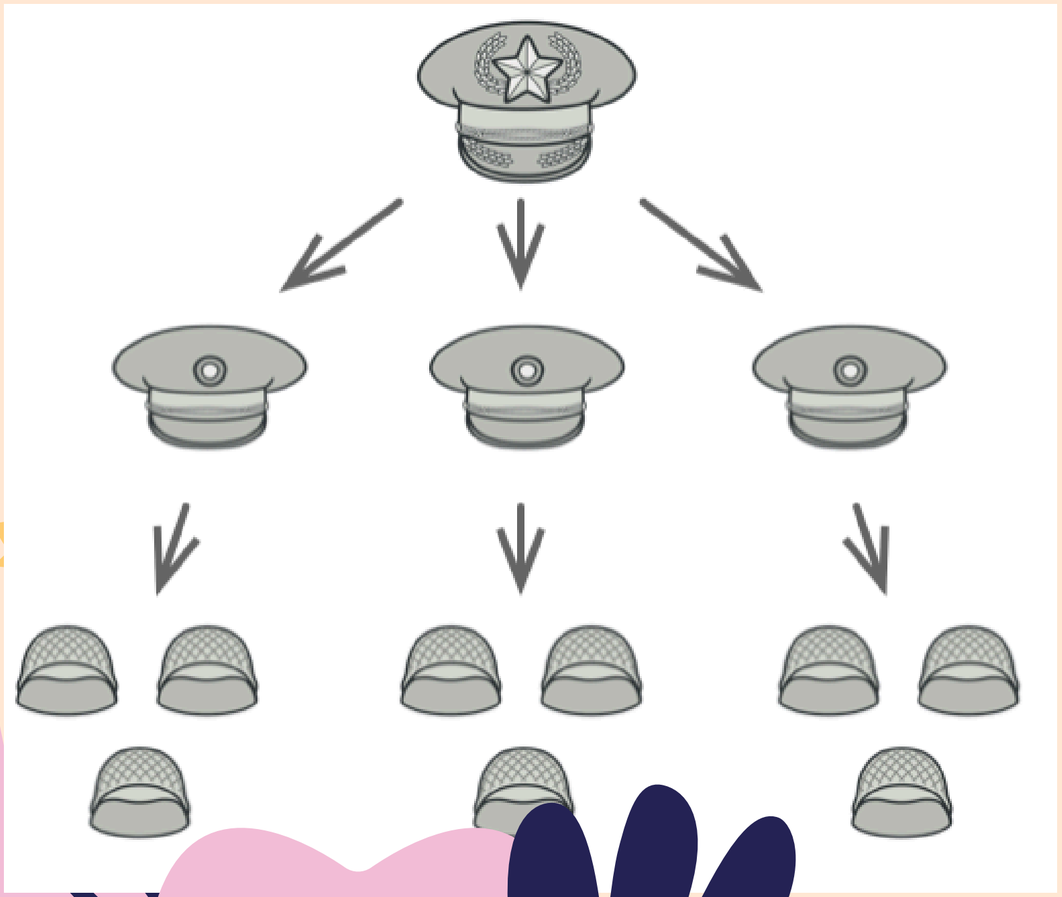
Composite Pattern

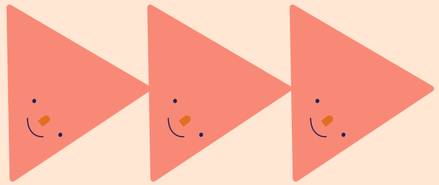
The common solution?

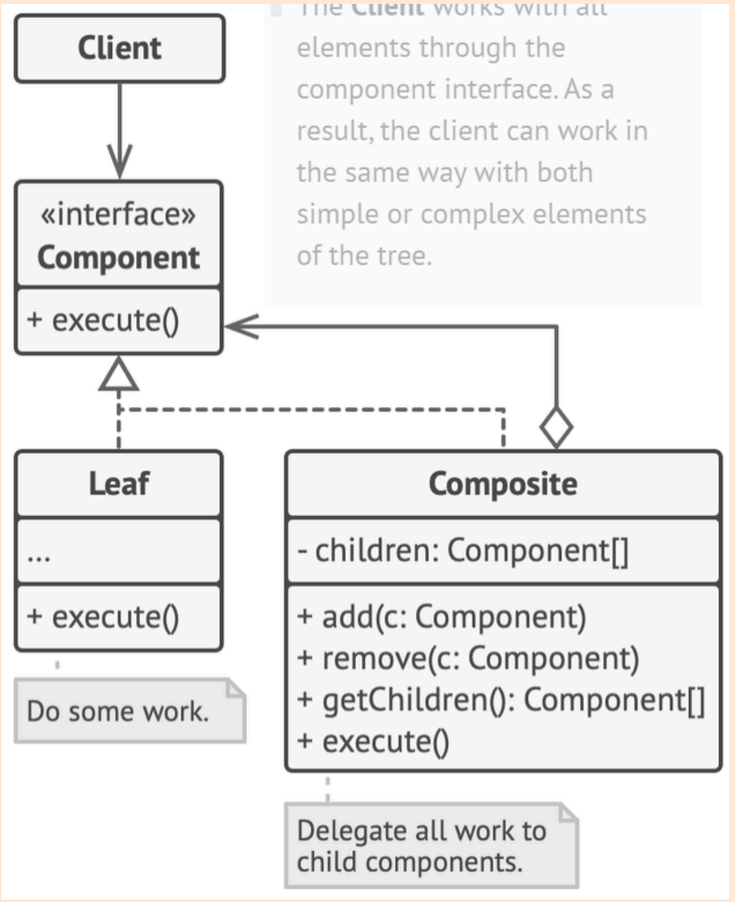
1. Identify leaves and composites
2. Create Leaf and Composite classes that both implement a Component interface
3. Add a list of Components to the Composite class (since composites contain more composites or leaves):
   private Component[] children



Composite Pattern







Calculator







Factory Pattern

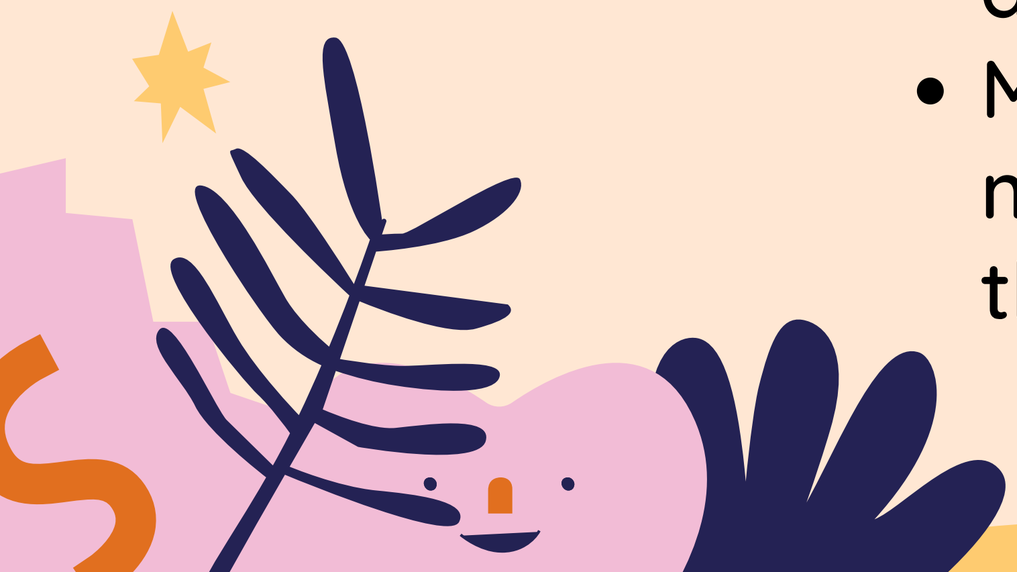


Factory Pattern

The common problem?

We want to create different types of objects depending on the conditions:

- Shipping company wants to create different vehicles (cargo ship, cargo aircraft, truck) depending on needs.
- Minecraft needs to spawn in different mobs depending on what conditions the player is in (creepers!!!).



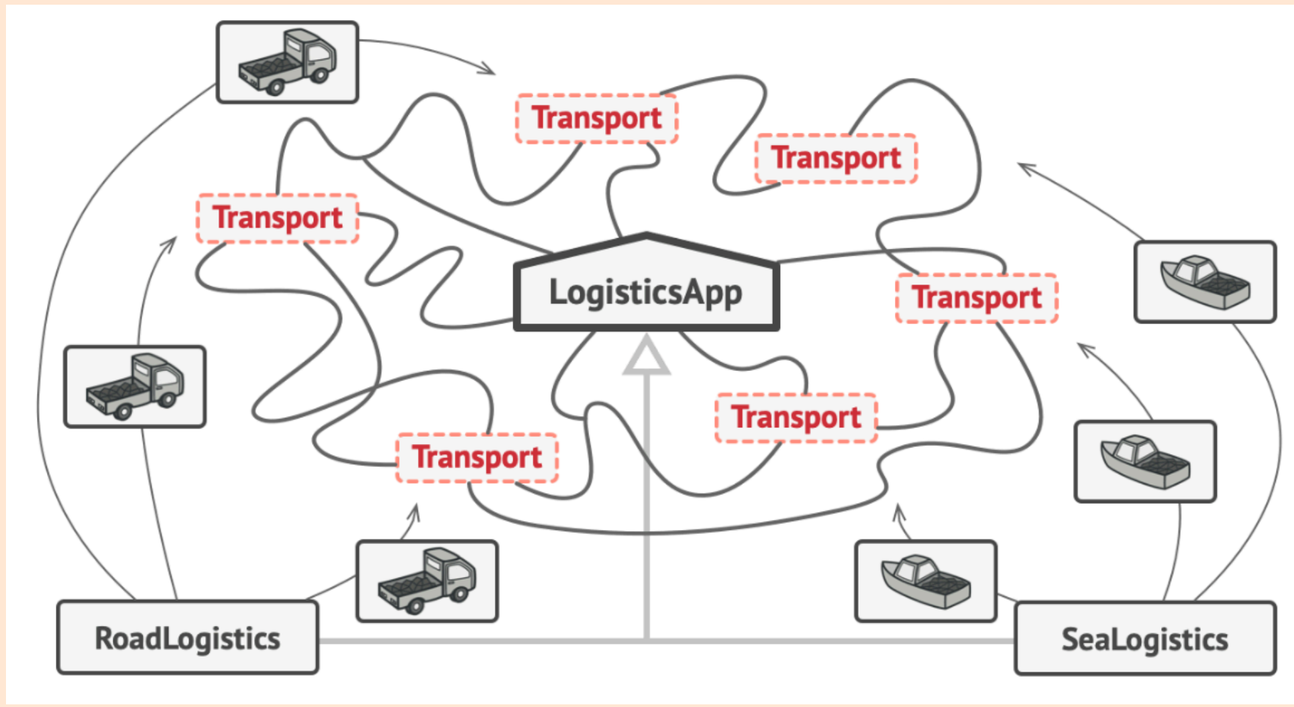
Factory Pattern

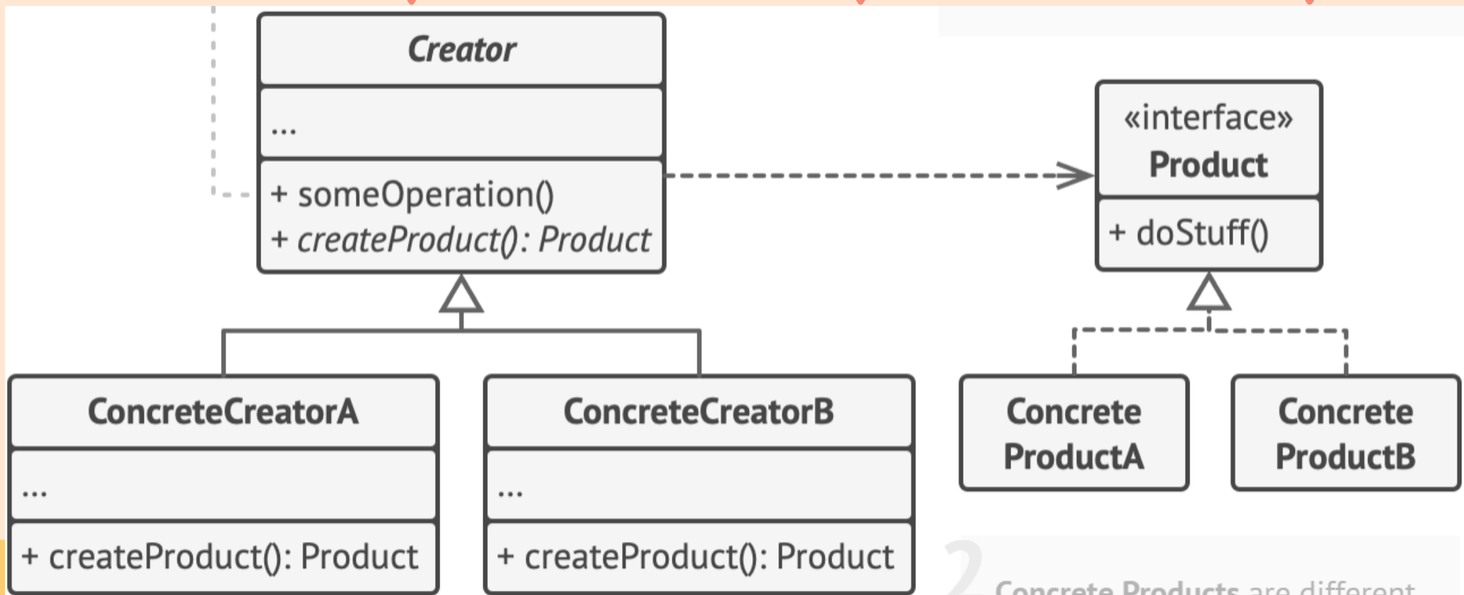
The common solution?

1. Create Factory abstract class, with a createProduct() abstract method.
2. Create Concrete Factories extending Factory for each Object.
3. Create a Product interface that Factory implements, with an action() abstract method.
4. Make each Object implement the Product interface



Factory Pattern







Factory Pattern & Abstraction

How does the Factory Pattern allow us to abstract object construction?



Thrones

